

1. (a) $f((a_1, a_2)) = \begin{cases} 4/36 & \text{se } (a_1, a_2) \in \{(0, 0), (1, 0)\} \\ 12/36 & \text{se } (a_1, a_2) \in \{(0, 1)\} \\ 9/36 & \text{se } (a_1, a_2) \in \{(0, 2)\} \\ 6/36 & \text{se } (a_1, a_2) \in \{(1, 1)\} \\ 1/36 & \text{se } (a_1, a_2) \in \{(2, 0)\} \\ 0 & \text{se c.c.} \end{cases} \quad \text{(b) } \frac{5}{36}$
- (c) $f_X(a_1) = \begin{cases} 25/36 & \text{se } a_1 = 0 \\ 10/36 & \text{se } a_1 = 1 \\ 1/36 & \text{se } a_1 = 2 \\ 0 & \text{se c.c.} \end{cases}, \quad f_Y(a_2) = \begin{cases} 9/36 & \text{se } a_2 \in \{0, 2\} \\ 18/36 & \text{se } a_2 = 1 \\ 0 & \text{se c.c.} \end{cases}$
- (d) Não (e) $Cov(X, Y) = -\frac{1}{6}, \quad \rho(X, Y) = -1/\sqrt{5}$.
2. (a) $f((a_1, a_2)) = \begin{cases} 1/16 & \text{se } (a_1, a_2) \in \{(0, 0), (0, 2), (1, 0), (1, 3), (2, 1), (2, 3)\} \\ 2/16 & \text{se } (a_1, a_2) \in \{(0, 1), (2, 2)\} \\ 3/16 & \text{se } (a_1, a_2) \in \{(1, 1), (1, 2)\} \\ 0 & \text{se c.c.} \end{cases} \quad \text{(b) } \frac{11}{16}$
- (c) $f_{X_1}(a_1) = \begin{cases} 4/16 & \text{se } a_1 \in \{0, 2\} \\ 8/16 & \text{se } a_1 = 1 \\ 0 & \text{se c.c.} \end{cases}; \quad f_{X_2}(a_2) = \begin{cases} 2/16 & \text{se } a_2 \in \{0, 3\} \\ 6/16 & \text{se } a_2 \in \{1, 2\} \\ 0 & \text{se c.c.} \end{cases}; \text{ Não}$
- (d) $Cov(X_1, X_2) = \frac{1}{4}, \quad \rho(X_1, X_2) = 1/\sqrt{6}$
3. (a) $C_{(X,Y)} = \{(0, 1), (1, 0), (1, 1), (2, 0), (2, 1), (3, 0), (3, 1)\};$
- $F((c_1, c_2)) = P(X \leq c_1, Y \leq c_2) = \begin{cases} 0 & \text{se } (c_1 < 0 \vee c_2 < 0) \vee (0 \leq c_1 < 1 \wedge 0 \leq c_2 < 1) \\ 1/32 & \text{se } 1 \leq c_1 < 2 \wedge 0 \leq c_2 < 1 \\ 5/32 & \text{se } 2 \leq c_1 < 3 \wedge 0 \leq c_2 < 1 \\ 14/32 & \text{se } c_1 \geq 3 \wedge 0 \leq c_2 < 1 \\ 1/32 & \text{se } 0 \leq c_1 < 1 \wedge c_2 \geq 1 \\ 4/32 & \text{se } 1 \leq c_1 < 2 \wedge c_2 \geq 1 \\ 13/32 & \text{se } 2 \leq c_1 < 3 \wedge c_2 \geq 1 \\ 1 & \text{se } c_1 \geq 3 \wedge c_2 \geq 1 \end{cases}$
- (b) $f_X(x) = \begin{cases} 1/32 & \text{se } x = 0 \\ 3/32 & \text{se } x = 1 \\ 9/32 & \text{se } x = 2 \\ 19/32 & \text{se } x = 3 \\ 0 & \text{se c.c.} \end{cases}; \quad f_Y(y) = \begin{cases} 14/32 & \text{se } y = 0 \\ 18/32 & \text{se } y = 1 \\ 0 & \text{se c.c.} \end{cases}; \text{ Não}$
- (c) $\frac{78}{32}, \frac{18}{32}; 0.621; 0.246; -0.059; -0.151$
4. $Cov(X, X^2) = \rho(X, X^2) = 0$.
5. (a) $F_M(c) = [F(c)]^n; F_N(c) = 1 - [1 - F(c)]^n$ (b) $Exp(n\lambda)$
6. São independentes.
7. $X_1 + X_2 + \dots + X_n \sim Bin(n, p)$

8. (a) $f_X(x) = \begin{cases} 0 & \text{se } x \leq 0 \\ e^{-x} & \text{se } x > 0 \end{cases}$; $f_Y(y) = \begin{cases} 0 & \text{se } y \leq 0 \\ e^{-y} & \text{se } y > 0 \end{cases}$ (b) $\frac{1}{2}$
- (c) $P(X+Y \leq u) = \begin{cases} 0 & \text{se } u \leq 0 \\ 1 - e^{-u}[1+u] & \text{se } u > 0 \end{cases}$; $2e^{-1} - 3e^{-2}$ (d) São independentes
- (e) $E[X] = E[Y] = 1$, $Var[X] = Var[Y] = 1$, $Cov(X, Y) = \rho(X, Y) = 0$
9. (a) $k = 1/8$; $f_X(x) = \begin{cases} \frac{1}{4}x^3 & \text{se } 0 < x < 2 \\ 0 & \text{se } c.c. \end{cases}$; $f_Y(y) = \begin{cases} \frac{1}{8}[\frac{8}{3} - 2y + \frac{5}{6}y^3] & \text{se } -2 < y \leq 0 \\ \frac{1}{8}[\frac{8}{3} - 2y + \frac{1}{6}y^3] & \text{se } 0 < y < 2 \\ 0 & \text{se } c.c. \end{cases}$
- (b) Não (c) $\frac{8}{5}$; $-\frac{8}{15}$; 0.107; 0.604; -0.036 ; -0.142
10. (a) $f_X(x) = \begin{cases} 0 & \text{se } x < 0 \vee x > 1 \\ \frac{2}{5}(x+2) & \text{se } 0 \leq x \leq 1 \end{cases}$; $f_Y(y) = \begin{cases} 0 & \text{se } y < 0 \vee y > 1 \\ \frac{1}{5}(1+8y) & \text{se } 0 \leq y \leq 1 \end{cases}$;
Não são independentes
- (b) $\frac{8}{15}$ (c) $\frac{2}{5}$ (d) $\frac{8}{15}$; $\frac{19}{30}$; $\frac{37}{450}$; $\frac{59}{900}$; $-\frac{1}{225}$; -0.061
11. 0.12
12. 0.0034
13. $(X, Y) \sim M(2; \frac{1}{6}, \frac{1}{2})$; Não
14. Note que $\det(\Sigma) = \sigma_1^2 \sigma_2^2 \dots \sigma_p^2 \neq 0$, pelo que a matriz Σ é invertível. Mais $\Sigma^{-1} = \text{Diagonal} \left(\frac{1}{\sigma_1^2}, \frac{1}{\sigma_2^2}, \dots, \frac{1}{\sigma_p^2} \right)$.
15. Note que $\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$, $\det(\Sigma) = \sigma_1^2 \sigma_2^2 (1-\rho^2) \neq 0$ e $\Sigma^{-1} = \frac{1}{\sigma_1^2 \sigma_2^2 (1-\rho^2)} \begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}$.
 $X_1 \sim N(u_1, \sigma_1^2)$ e $X_2 \sim N(u_2, \sigma_2^2)$.
16. (a) — (b) $Cov(X_1, X_2) = \rho(X_1, X_2) = \frac{2}{\pi}$ (c) —
17. (a) $\chi_{0.25} = 1.75$; $\chi_{0.5} = 2.5$; $\chi_{0.75} = 3.25$ (b) 0.0003
- (c) i. $f((x, y)) = \begin{cases} \frac{1}{9} & \text{se } x \in [1, 4], y \in [1, 4] \\ 0 & \text{se } c.c. \end{cases}$ ii. $\frac{1}{2}$ iii. $\frac{4}{9}$

Soluções da Folha Prática 9

1. —
2. (a) $N(270, 67)$ (b) 0.0334
3. 0.0277
4. —
5. (a) 0.9997154 (b) 0.9997965 (c) igual a (b)
6. $n = 62$. Note que se quer determinar $n \in \mathbb{N}$ tal que $P(|\bar{X}_n - \mu| \leq 0.25\sigma) \geq 0.95$, em que μ e σ são, respetivamente, o valor médio e o desvio-padrão da v.a.r. e $\bar{X}_n$ denota a média amostral.
7. —
8. —
9. (a) $E[Y] = 350$; $Var[Y] = \frac{875}{3}$; 0.0716
(b) $X \sim Bin(100, \frac{1}{6})$; $E[X] = \frac{100}{6}$; $Var[X] = \frac{500}{36}$; 0.9998267 (aproximado); 0.9997042 (exato)
(c) Não são independentes